

4. Select a particular example and right-click the file to run it.

The deep neural networks are made up of multiple layers. The `MultiLayerConfiguration` with parameters is customised to suit the requirements of the current problem. The hyperparameter variable decides the learning behaviour of the network. These parameters include the following:

- Number of times to update the weights of the model
- Initialisation mechanism for these weights
- The type of activation function to link with the nodes
- The specific optimisation algorithm to be used
- The speed with which the network should learn

A sample configuration is shown in the following code snippet:

```
MultiLayerConfiguration conf = new NeuralNetConfiguration.  
Builder()  
    .iterations(1)  
    .weightInit(WeightInit.XAVIER)  
    .activation("relu")  
    .optimizationAlgo(OptimizationAlgorithm.STOCHASTIC_  
GRADIENT_DESCENT)  
    .learningRate(0.05)  
    // ... other hyperparameters  
    .list()  
    .backprop(true)  
    .build();
```

A new layer can be added by invoking `layer()` on `NeuralNetConfiguration.Builder()`. In this process, we need to specify the following:

- Location (order) where the layer has to be added
- The number of input and output nodes (nIn and nOut)
- The type of layer

```
layer(0, new DenseLayer.Builder().nIn(784).nOut(250)  
    .build());
```

After completing the configuration process, the training of the model can be carried out with the following command:

```
model.fit
```

DL4J's neural networks

DL4J supports many powerful neural networks. Some of them are listed below:

- Restricted Boltzmann machines
- Convolutional nets
- Recurrent nets
- Deep-belief networks
- Stacked denoising autoencoders

The reason for choosing JVM

Most of the deep learning/machine learning libraries are in

Python. But DL4J is based on JVM. One may ask why this deliberate choice was made. The official documentation lists the major reasons for choosing the Java Platform:

- Many large government organisations and companies have Java as their major platform. There are millions of Java developers and Java is the largest programming language, as of now.
- Java is the basis of many powerful techniques/tools such as Hadoop, Elasticsearch, Lucene and Hive.
- One prime complaint against Python is its slow speed. Java is definitely quicker than Python. Hence, when handling projects with massive data sets, Java may be a better choice.
- Java is inherently secure and cross-platform. It can be easily used on all major platforms such as Linux, Windows, OSX and Android.

DL4J on Android

Deeplearning4j can be used with Android mobile phones as well. The prerequisites for running it on Android are:

- Emulator (with API level 21 or above) or a device
- Android Studio 2.2 or later

The following dependencies must be added to the *build.gradle* file:

```
compile 'org.deeplearning4j:deeplearning4j-core:0.8.0'  
compile 'org.nd4j:nd4j-native:0.8.0'  
compile 'org.nd4j:nd4j-native:0.8.0:android-x86'  
compile 'org.nd4j:nd4j-native:0.8.0:android-arm'  
compile 'org.bytedeco.javacpp-presets:openblas:0.2.19-  
1.3:android-x86'  
  
compile 'org.bytedeco.javacpp-presets:openblas:0.2.19-  
1.3:android-arm'
```

A clean and easy-to-follow tutorial is available at <https://deeplearning4j.org/android>.

This article has provided an introduction to the Deeplearning4J library. If you are interested in exploring more about deep learning, there are many interesting online courses and resources. A compiled list of links is available at <https://deeplearning4j.org/deeplearningforbeginners.html>. 

References

- [1] Deeplearning4j official homepage: <https://deeplearning4j.org/>
- [2] Maven: <http://maven.apache.org/guides/getting-started/maven-in-five-minutes.html>

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